

compute descriptive statistics for a real dataset; identify skewed distributions; apply zscore normalization; compute pearson and spearman correlation

```
In [17]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load Titanic dataset
df = sns.load_dataset('tips')
#df = pd.read_json("titanic.json")

# Descriptive statistics
stats = df.describe()

print(stats)
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
In [18]: # Select numeric columns
num_cols = df.select_dtypes(include='number')

# Calculate skewness
skewness = num_cols.skew()
print("Skewness of Numeric Columns:")
print(skewness)
```

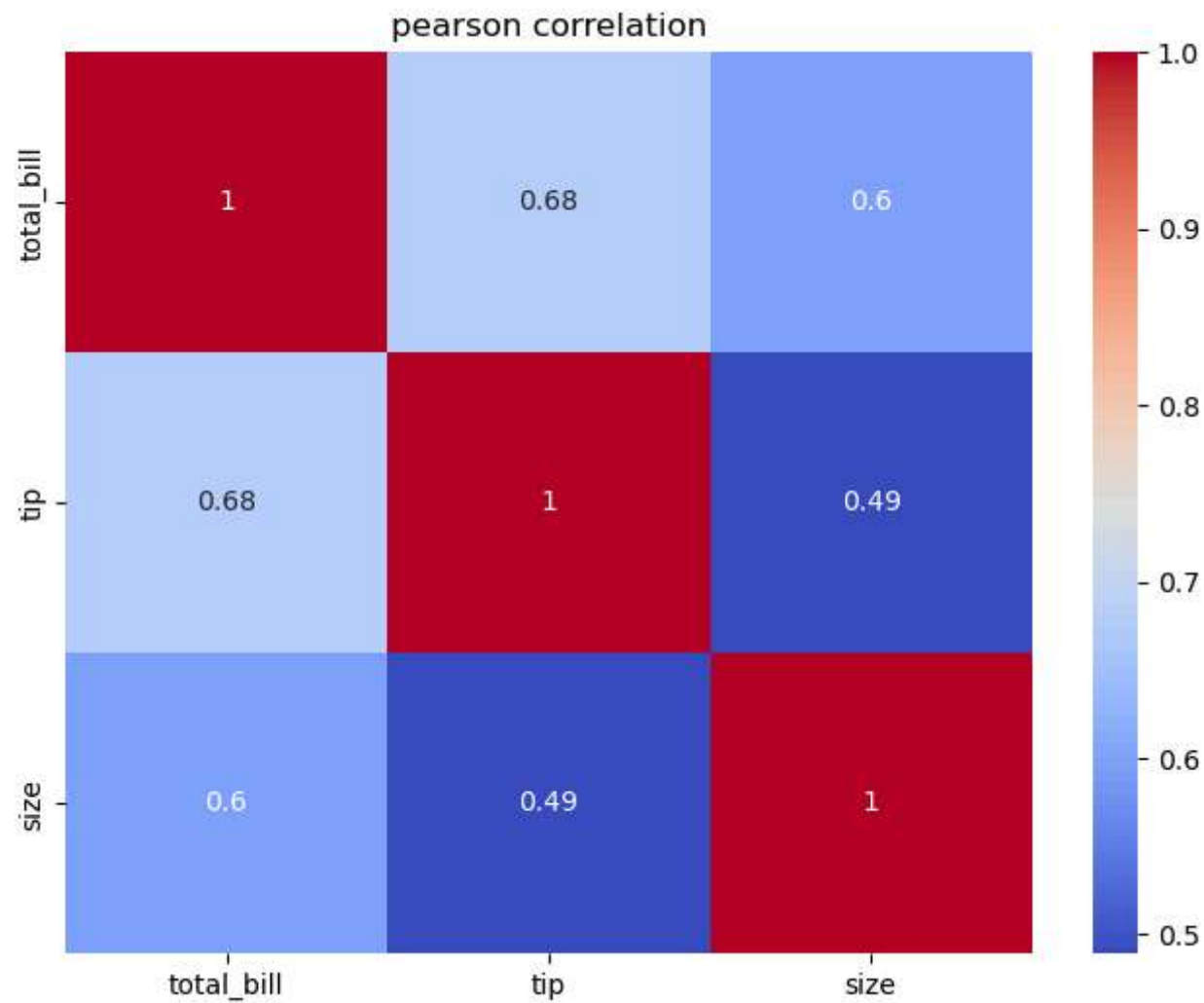
```
Skewness of Numeric Columns:
total_bill    1.133213
tip           1.465451
size          1.447882
dtype: float64
```

```
In [22]: # Apply Z-score normalization
zscore_df = num_cols.copy()
for col in zscore_df.columns:
    zscore_df[col] = (
        zscore_df[col] - zscore_df[col].mean()
    ) / zscore_df[col].std()
print(zscore_df.head())
```

	total_bill	tip	size
0	-0.314066	-1.436993	-0.598961
1	-1.061054	-0.967217	0.452453
2	0.137497	0.362610	0.452453
3	0.437416	0.225291	-0.598961
4	0.539635	0.442111	1.503867

```
In [26]: # Pearson Correlation
pearson_corr = num_cols.corr(method='pearson')
print(pearson_corr)
#Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(
    pearson_corr,
    annot=True,
    cmap='coolwarm'
)
plt.title("pearson correlation")
plt.show()
```

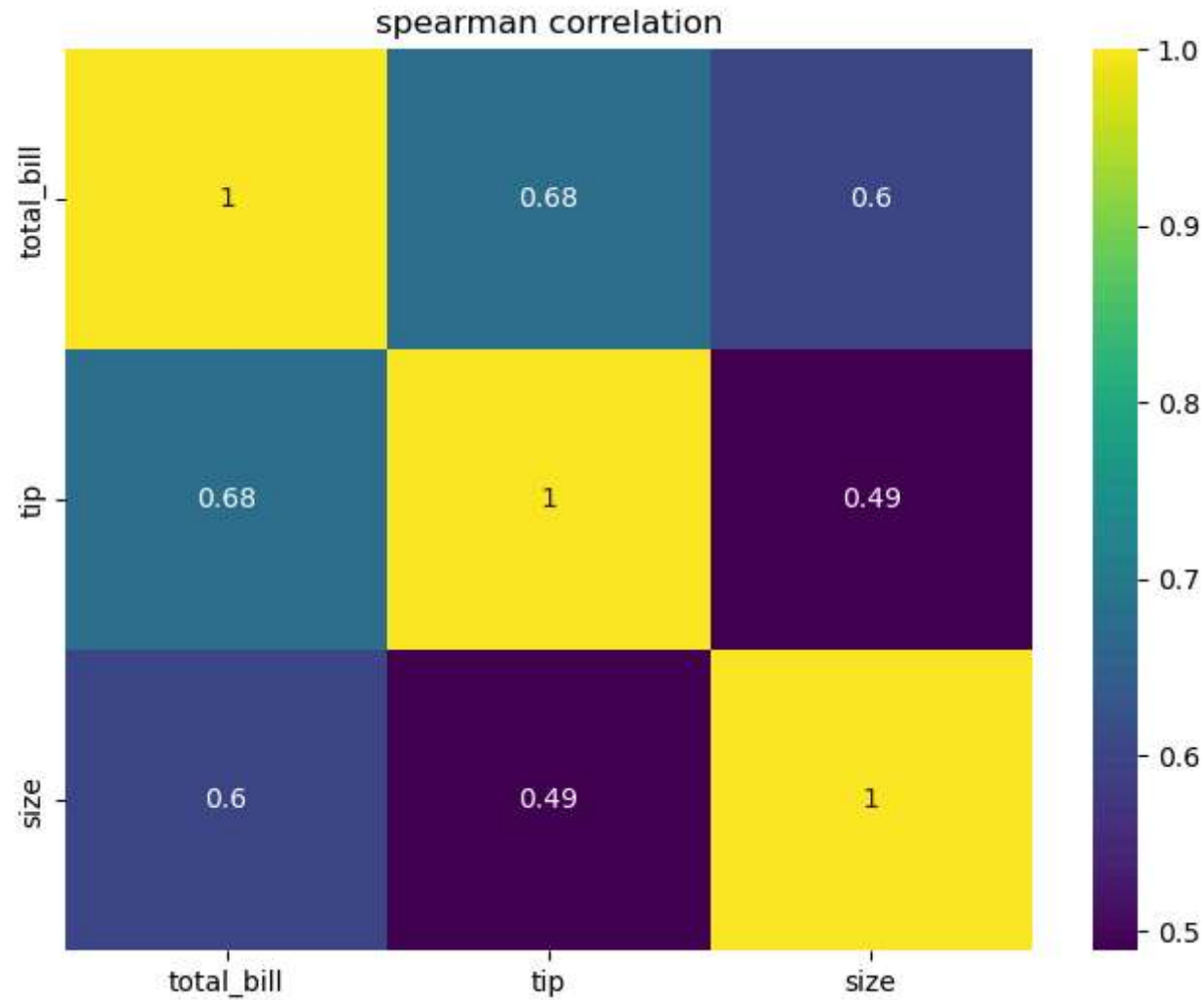
	total_bill	tip	size
total_bill	1.000000	0.675734	0.598315
tip	0.675734	1.000000	0.489299
size	0.598315	0.489299	1.000000



```
In [29]: # Spearman Correlation
spearman_corr = num_cols.corr(method='spearman')
print(spearman_corr)
#Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(
    pearson_corr,
    annot=True,
    cmap='viridis'
)
```

```
plt.title("spearman correlation")
plt.show()
```

	total_bill	tip	size
total_bill	1.000000	0.678968	0.604791
tip	0.678968	1.000000	0.468268
size	0.604791	0.468268	1.000000



```
In [38]: #compare pearson and spearman
comparison = pd.DataFrame({
    "pearson": pearson_corr["total_bill"],
```

```
    "spearsman": spearman_corr["total_bill"]  
  })  
  print(comparison)
```

	pearson	spearsman
total_bill	1.000000	1.000000
tip	0.675734	0.678968
size	0.598315	0.604791